Compressibility: Adversarial Robustness and Generalization

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https://github.com/ChafikHala/netcompression

I. INTRODUCTION

Deep neural networks are often trained with heavily overparameterized architectures, yet a large body of work has shown that trained models can frequently be compressed to a fraction of their original size through pruning or other parameter reduction techniques [2], [3]. This suggests that predictors learned by large networks may admit significantly more compact representations than the models used during training. Understanding this gap between model size and effective complexity has therefore become an important question in modern deep learning.

Beyond accuracy, compression may also affect other properties of the learned predictor. Recent work [1] suggests that structured compression can influence adversarial robustness. When predictive information becomes concentrated in fewer directions or parameters, the resulting representations may become more sensitive to carefully crafted perturbations. In this work, we study **adversarial weakness under structured compressibility using neuron and spectral compression**.

Compressibility also raises questions about the role of overparameterization in optimization. If a predictor obtained with a large network can ultimately be represented with far fewer parameters, it is unclear whether large models are necessary because the solution itself is complex, or because their parameterization facilitates optimization. To investigate this, we compare predictors obtained by compressing large trained networks with **small dense models of comparable parameter budgets trained directly from scratch** [4].

Finally, compressibility may also shed light on generalization. To explore this, we perform experiments with varying levels of **label noise**. While sufficiently large networks can fit random labels, such memorization may lead to solutions that are less compressible. Studying how compressibility evolves under increasing label noise may therefore help distinguish solutions that capture meaningful structure from those that primarily memorize the data.

Understanding these phenomena has both theoretical and practical implications. Insights into compressibility can guide the design of more efficient models with lower computational and memory requirements, enabling deployment on resource-constrained devices. Also, analyzing how compression affects robustness contributes to the broader goal of developing reliable and secure AI systems.

II. ADVERSARIAL ROBUSTNESS

Background. Several previous works have studied the effect of compressibility on adversarial robustness. In this report, we mainly adopt the framework of [1], which argues that structured compressibility can have a detrimental effect on robustness by increasing the sensitivity of the network to adversarial perturbations. More precisely, the paper distinguishes between *neuron compressibility*, where a small number of rows dominate weight matrices, and *spectral compressibility*, where a small number of singular values dominate. The main theoretical idea is that these forms of compressibility increase layer operator norms, which in turn increase the Lipschitz constant of the network and therefore its vulnerability to adversarial attacks. A central result of the paper is Theorem *3.1*, which relates structured compressibility to operator norms. In the neuron-compressible case, if $v := (\|w_1\|_1, \dots, \|w_h\|_1)$ is the vector of row norms of a layer matrix W and is $(1, k_v, \epsilon_v)$-compressible[1], and if each row is itself $(2, k_r, \epsilon_r)$-compressible, then

$$\|W\|_\infty \leq \frac{1-\epsilon_v}{1-\beta_v}\left(\frac{\sqrt{h}\,k_r + h\epsilon_r}{k_v}\right)\|W\|_F. \tag{4}$$

Similarly, in the spectrally compressible case, if the singular value vector $\sigma := (\sigma_1, \sigma_2, \dots)$ is $(1, k_\sigma, \epsilon_\sigma)$-compressible, then

$$\|W\|_2 \leq \frac{1-\epsilon_\sigma}{1-\beta_\sigma}\left(\sqrt{\frac{h}{k_\sigma}}\right)\|W\|_F. \tag{5}$$

For a fixed Frobenius norm, increasing compressibility increases the upper bounds on the relevant operator norms of the network layers. Since adversarial robustness can be controlled through Lipschitz-type bounds that depend on the product of these operator norms across layers (up to interlayer alignment terms), larger operator norms correspond to greater sensitivity of the network to input perturbations. The framework therefore predicts a negative relationship between compressibility and adversarial robustness. We now investigate how these theoretical insights are borne out empirically.

Methodology. We train a one-layer fully connected network with 400 hidden units and ReLU activations on the Binary MNIST dataset. During training, we incorporate

[1] Given $\theta \in \mathbb{R}^d$ and a non-negative integer $k \leq d$, let θ_k denote the vector retaining the k largest-magnitude elements of θ with all others set to zero. Then θ is (q, k, ϵ)-compressible if and only if $\|\theta - \theta_k\|_q / \|\theta\|_q \leq \epsilon$.

a nuclear norm regularization term (enforcing spectral compressibility) $\|W\|_*$ weighted by a hyperparameter α. Additionally, to avoid confounding from nuclear norm regularization decreasing the overall parameter norms, the Frobenius norm of the weight matrix is normalized to its initial value at each training iteration.

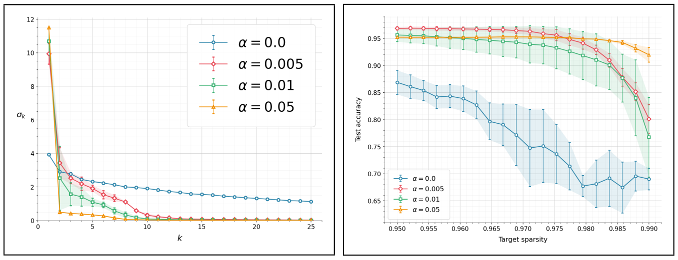

Fig. 1. Effect of spectral regularization α on model compressibility and pruning robustness. **Left**: singular value spectra of the trained weight matrices. **Right**: test accuracy under increasing target sparsity.

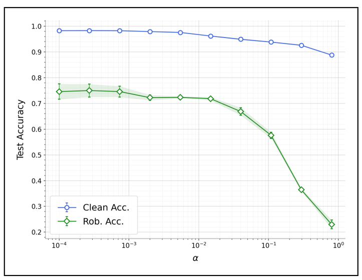

Fig. 2. Standard and adversarial test accuracy as a function of the regularization strength α.

Fig. 1 confirms the expected effect of nuclear norm regularization: as α increases, the singular value spectrum decays more rapidly (left panel). The right panel reveals that models trained with larger α maintain higher test accuracy under pruning, whereas the unregularized model ($\alpha = 0$) suffers a much steeper degradation in performance as sparsity increases.

Fig. 2 further shows that greater spectral compressibility comes at a cost: adversarial accuracy falls sharply as α increases under an ℓ_2-FGSM attack.

Next, we train a four-layer FCN with 2000 hidden units and a WideResNet-16-2 on CIFAR-10. To enforce neuron compressibility, we use Group Lasso regularization for the FCN, which induces row sparsity through a regularization parameter β. For the WideResNet-16-2, we instead use the scale-invariant ℓ_1/ℓ_2 penalty, in order to avoid confounding effects due to changes in the overall parameter scale and to make the regularization act on row sparsity rather than row magnitude. Spectral compressibility is enforced via low-rank approximation. We measure adversarial accuracy—using ℓ_∞-FGSM for neuron compressibility and ℓ_2-FGSM for spectral compressibility—as well as the relative representation shift $\frac{\|z_{\text{adv}}-z\|_2}{\|z\|_2}$, where $z = \Phi(x)$ and $z_{\text{adv}} = \Phi(x + a^*)$ denote the learned representations of a clean and adversarially perturbed input, respectively, and a^* is the worst-case perturbation (Φ is the network under study without its classification head).

Figures 3–5 illustrate how different compression constraints affect performance and adversarial sensitivity across architectures. In Fig. 3, increasing the Group Lasso regularization strength β progressively reduces adversarial accuracy and the robust-to-clean accuracy ratio, while the representation shift increases, indicating greater sensitivity to adversarial perturbations. Fig. 4 shows that relaxing the low-rank constraint improves adversarial accuracy and reduces the representation shift, suggesting more stable internal representations. Fig. 5 confirms these patterns on WideResNet-16-2, with stronger row sparsity degrading robustness and higher rank improving adversarial performance.

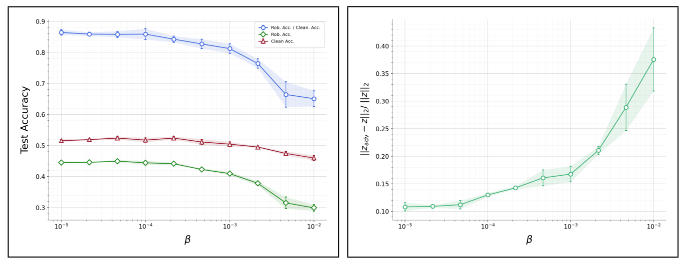

Fig. 3. Effect of group Lasso regularization strength β on model performance and adversarial sensitivity for the FCN trained on CIFAR-10. **Left**: clean accuracy, adversarial accuracy, and their ratio as a function of β. **Right**: relative representation shift $\|z_{\text{adv}} - z\|_2 / \|z\|_2$.

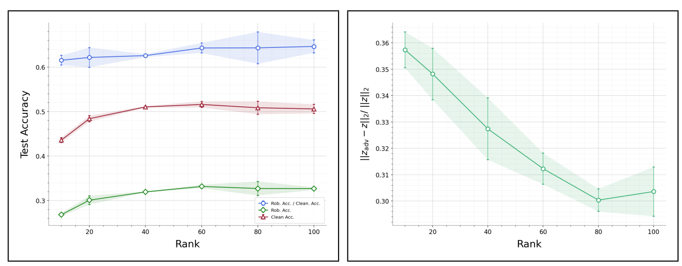

Fig. 4. Effect of low-rank approximation on model performance and representation sensitivity for the FCN trained on CIFAR-10. **Left**: clean accuracy, adversarial accuracy, and their ratio as a function of the rank constraint. **Right**: relative representation shift $\|z_{\text{adv}} - z\|_2 / \|z\|_2$

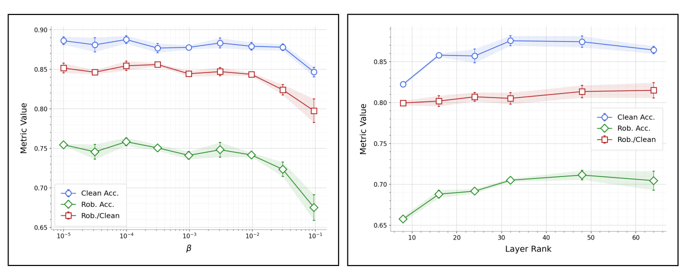

Fig. 5. Clean accuracy, adversarial accuracy, and their ratio as a function of the compression constraint for WideResNet-16-2 trained on CIFAR-10. **Left**: effect of the Group Lasso regularization strength β controlling row sparsity. **Right**: effect of the layer rank under low-rank approximation.

Fig.6 shows how adversarial and universal adversarial robustness respond to changes in Frobenius norm and nuclear norm regularization α. Increasing the Frobenius norm (left) sharply reduces standard adversarial robustness (the Rob./Clean ratio falls from 0.92 to 0.62) while the UAE/Clean ratio remains relatively stable, declining only marginally from 0.98 to 0.94. This asymmetry suggests that Frobenius norm scaling primarily affects vulnerability to input-specific perturbations, without substantially increasing susceptibility to universal adversarial examples. Increasing α (right) produces a qualitatively different pattern: both robustness measures deteriorate in tandem, with the Rob./Clean ratio dropping from 0.60 to 0.25 and the UAE/Clean ratio from 0.90 to 0.72. This indicates that nuclear norm regularization renders the model vulnerable not only to adversarial perturbations but also to universal adversarial examples, pointing to a more fundamental degradation of the learned representations.

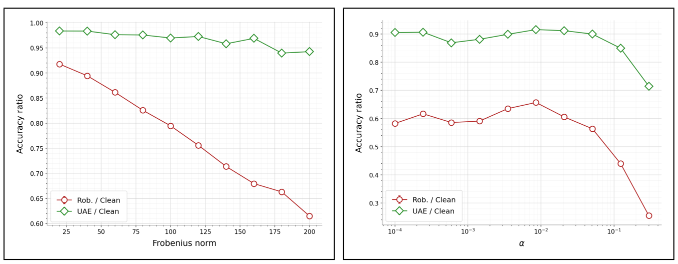

Fig. 6. Vulnerability to universal adversarial examples (UAE) on binary MNIST for a one-layer FCN. **Left**: effect of the Frobenius norm of the weight matrix. **Right**: effect of the spectral regularization strength α.

III. OVERPARAMETRIZATION AND OPTIMIZATION

Methodology. To investigate the role of overparameterization in optimization, we train MobileNet models on CIFAR-10 with different capacity levels controlled by the width multiplier parameter. Specifically, we train dense models with width multipliers 0.25, 0.5, 0.75, and 1.0, each evaluated across five random seeds.

In addition, starting from the full-width model (1.0), we apply gradual global magnitude pruning during training. In this procedure, weights are progressively removed according to their magnitude across the entire network: at regular intervals during training, a fraction of the smallest-magnitude weights is set to zero globally rather than layer by layer. The sparsity level is increased following an epoch-based linear schedule, starting from an initial sparsity of 0 and gradually reaching a target sparsity. We consider target sparsity levels of 45%, 75%, and 95%, and repeat each experiment across five seeds.

The training loss curves (Fig. 7) are smooth and very similar across models, indicating stable optimization for both dense and pruned networks. Differences appear more clearly in the validation loss: smaller dense models (especially width 0.25) converge to noticeably higher loss values, reflecting limited capacity. In contrast, the sparse models obtained from the full-width network reach validation losses close to those of the large dense model, showing that even at high sparsity levels the pruned networks retain training dynamics and generalization behavior comparable to the original model.

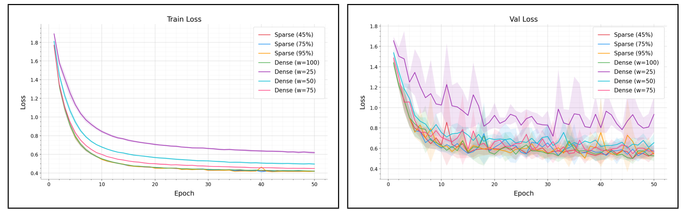

Fig. 7. Training dynamics of MobileNet models on CIFAR-10 for dense networks with varying width multipliers and sparse networks obtained via gradual global magnitude pruning. **Left**: training loss across training epochs. **Right**: validation loss across training epochs.

Model	Non-zero Params	Top-1 Acc.	Top-5 Acc.
Dense 0.25	221,141	0.7645 ± 0.0102	0.9859 ± 0.0015
Sparse 95%	203,579	**0.8190 ± 0.0086**	**0.9902 ± 0.0016**
Dense 0.50	834,693	0.8128 ± 0.0078	**0.9908 ± 0.0015**
Sparse 75%	842,645	0.8107 ± 0.0181	0.9901 ± 0.0024
Dense 0.75	1,840,693	**0.8319 ± 0.0045**	**0.9903 ± 0.0006**
Sparse 45%	1,801,243	0.8129 ± 0.0106	0.9898 ± 0.0015
Dense 1.00	3,239,141	0.8434 ± 0.0105	0.9920 ± 0.0016

TABLE I
MOBILENET PERFORMANCE ON CIFAR-10.

The results of Table I highlight that sparse models obtained by pruning a large network can remain highly competitive with dense models trained at comparable parameter budgets. In the smallest budget regime, the sparse 95% model substantially outperforms the dense 0.25 network despite using a similar number of parameters. At intermediate budgets (dense 0.50 vs. sparse 75%), the performance is nearly identical. However, for larger parameter counts, the dense 0.75 model slightly outperforms the sparse 45% network.

Overall, this experiment supports (particularly in the extreme compression regime) the hypothesis that starting from a larger, overparameterized model facilitates a more effective exploration of the loss landscape during training. The larger network provides sufficient capacity for optimization to reach well-performing regions of the parameter space, after which pruning can remove a large fraction of weights while retaining much of the performance. In contrast, training a small dense model of the same final size restricts the optimization trajectory from the outset, making it harder to reach similarly good solutions.

IV. OVERFITTING EFFECT ON COMPRESSIBILITY

Previous work proposed compression-based generalization bounds [5], suggesting that models that are more compressible tend to generalize better. Inspired by this connection, we investigate whether models that rely more heavily on memorization, and consequently exhibit stronger overfitting, become more fragile under post-hoc compression.

To this end, we train a WideResNet-16-2 on progressively noisier versions of the CIFAR-10 training set in order to encourage overfitting and memorization. For each noise level $\alpha \in [0,1]$, we construct a corrupted training set

$$D_\alpha = \{(x_i, y_i^{(\alpha)})\}_{i=1}^{n}$$

by randomly permuting the labels of an α-fraction of the training samples, while leaving the remaining labels unchanged. Thus, $\alpha = 0$ corresponds to the original dataset, whereas $\alpha = 1$ corresponds to fully shuffled labels. All models are trained independently for 100 epochs with SGD, Nesterov momentum, initial learning rate 0.1, zero weight decay, and a cosine annealing schedule. Inputs are normalized, and no data augmentation is used. We deliberately avoid introducing regularization mechanisms in order to avoid mixing their effects with those of label-noise injections.

For a model f_α trained on D_α, the training accuracy train_acc_α is evaluated on the same noisy training set. After training, each model is pruned by global magnitude pruning. For a pruning ratio p, we zero out the smallest fraction p of all convolutional and linear weights in absolute value. We then evaluate the pruned model on the same noisy training set D_α to obtain the pruned training accuracy $\text{train\_acc\_pruned}^{(\alpha,p)}$. We define the retained training accuracy by

$$\text{retained\_train\_acc}^{(\alpha,p)} = \frac{\text{train\_acc\_pruned}^{(\alpha,p)}}{\text{train\_acc}_\alpha}.$$

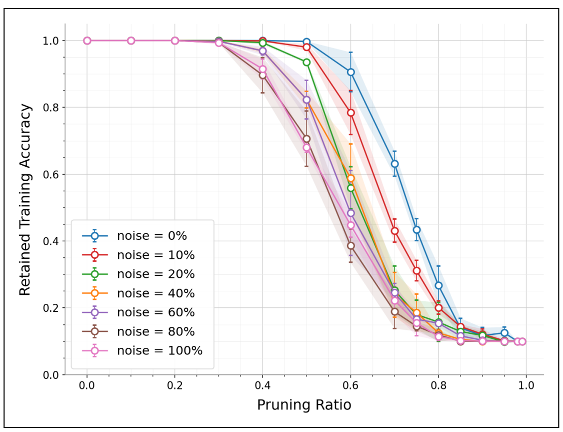

Fig. 8. Retained training accuracy as a function of the pruning ratio.

Fig. 8 plots $\text{retained\_train\_acc}^{(\alpha,p)}$ as a function of the pruning ratio p for all noise levels α, with error bars corresponding to ± 1 standard deviation across 3 seeds. As one would expect, the figure shows a clear monotonic trend: as the label-noise level increases, the retained training accuracy drops earlier. In particular, the model trained on clean labels (0% noise) remains stable up to substantially higher pruning ratios, whereas models trained with high noise levels (80% and 100%) begin to lose a large fraction of their training accuracy much sooner. This suggests that solutions obtained in regimes with stronger memorization and overfitting are more fragile under post-hoc compression.

To support this interpretation, we evaluate the overfitting rate

$$\text{overfit\_rate}_\alpha = \frac{\text{train\_acc}_\alpha - \text{test\_acc}_\alpha}{\text{train\_acc}_\alpha},$$

where test_acc_α denotes the standard accuracy on the clean held-out test set. We find that the overfitting rate increases monotonically with α. Moreover, due to the absence of regularization, all models fit their noisy training set almost perfectly within 100 epochs, including the model trained with fully shuffled labels. This indicates that, at high noise levels, training relies strongly on memorization.

This experiment suggests that models that rely more heavily on memorization tend to learn representations that are harder to simplify, whereas models that generalize better tend to learn simpler solutions that admit more compact representations. This is consistent with the idea that compressibility is linked not only to parameter efficiency, but also to the nature of the learned solution.

V. CONCLUSION

This report investigated three interconnected questions about neural network compressibility: its relationship to adversarial robustness, the role of overparameterization in optimization, and its connection to memorization and generalization.

Our experiments support the theoretical prediction of [1] that structured compressibility can negatively affect adversarial robustness. Across architectures and compression mechanisms, greater compression leads to increased adversarial sensitivity, higher representation shift under attack, and lower robust-to-clean accuracy ratios.

Regarding the role of overparameterization, our comparison between large sparse models and small dense models with comparable parameter budgets suggests that starting from an overparameterized network may enable a more effective exploration of the loss landscape. This effect is most visible in the extreme compression regime, where the pruned network significantly outperforms a dense model of similar size.

Finally, the label-noise experiments reveal that models trained in high-memorization regimes produce representations that are more fragile under post-hoc compression, as measured by retained training accuracy under pruning. This is consistent with the view that compressibility reflects not merely parameter efficiency, but the intrinsic simplicity of the learned solution.

Taken together, these results highlight that compressibility is a multi-faceted property of neural networks with implications beyond model efficiency, influencing adversarial vulnerability, optimization dynamics, and the structure of learned representations.

Limitations and Future Work. Several limitations should be noted. First, our experiments focus on relatively small-scale datasets and architectures, and it remains to be seen whether the same trends hold for modern large-scale models. Moreover, the analysis is primarily empirical, and further theoretical work would be needed to better explain the mechanisms linking compressibility robustness and generalization. Future work could therefore extend these experiments to larger models, explore alternative compression strategies, and investigate whether similar relationships emerge in other domains such as natural language processing or vision transformers.

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Justification: The introduction clearly states the three main objectives: analyzing the relationship between compressibility and adversarial robustness, studying the role of overparameterization in optimization, and investigating the connection between compressibility and generalization under label noise. These claims are directly supported by the empirical results presented in Sections II–IV and are appropriately scoped as primarily empirical.
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